DESCRIPTION OF COURSEWORK

Course Code	EEE 322
Course Name	Power Electronics
Lecturer	Ts. Dr. Kang Chia Chao
Academic Session	04-2024
Assessment Title	Assignment

A. Introduction/ Situation/ Background Information

Power electronics circuits convert electric power from one form to another using electronic device that operates as switches. An electronic switch is characterized by having the two states on and off, ideally being either a short circuit or an open circuit. Applications using switching devices are desirable because of the relatively small power loss in the device. If the switch is ideal, either the switch voltage or the switch current is zero, making the power absorbed by it zero. Real devices absorb some power when in the on state and when making transitions between the on and off states, but circuit efficiencies can still be quite high. Some electronic devices such as transistors can also operate in the active range where both voltage and current are nonzero, but it is desirable to use these devices as switches when processing power.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO1: Demonstrate the understanding in electrical power regulation in various power electronics applications.

CLO2: Analyze the models of the power converter control systems.

CLO 3: Develop power electronics solutions based on practical problems by applying the basic structures and principles of converters.

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:

   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).

   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.

   iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

- Group Assignment
- Copy-paste is prohibited and will significantly degrade your marks.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
CLO1: Demonstrate the understanding in electrical power regulation in various power electronics applications.		
1.	Introduction	10
2.	Overall Demonstration and Understanding	20
CLO2: Analyze the models of the power converter control systems.		
3.	Overall Content CLO 2	15
CLO3: Develop power electronics solutions based on practical problems by applying the basic structures and principles of converters.		
4.	Overall Content CLO 3	15
5.	Conclusion	20
6.	Appearance & Formatting	20
	TOTAL	**100**

F. Task

CHALLENGES OF POWER ELECTRONICS

Normally the personal computer will be using many DC-DC converters to power up a lot of components such as the memory load, Center Processing Unit or even HDD and etc. While the currents and voltages are not as large as in the previously discussed examples, the current densities can be very high and power loss has to be reduced to increase battery life. Due to that, many personal computer manufacturers are beginning to use wideband gap devices in their products. In the others hand, the biggest problem for power electronics devices is the widespread occurrence of fake components. Counterfeit substandard power electronics product or devices that purchase by the consumers is higher compare to low power semiconductor devices. This is because many power device suppliers purchase components from trading inventory, and counterfeit components are common and famous in such inventory. In summary, power electronics circuit designers face many challenges.

Objective	The objective of this study is to demonstrate and verifying current challenges of power electronics in semiconductor industry thus provide the solutions.
Problem Statement	Normally the personal computer will be using many DC-DC converters to power up a lot of components such as the memory load, center processing unit or even HDD and etc. While the currents and voltages are not as large as in the previously discussed examples, the current densities can be very high and power loss has to be reduced to increase battery life. Due to that, many personal computer manufacturers are beginning to use wideband gap devices in their products. In the others hand, the biggest problem for power electronics devices is the widespread occurrence of fake components. Counterfeit substandard power electronics product or devices that purchase by the consumers is higher compare to low power semiconductor devices. This is because many power device suppliers purchase components from trading inventory, and counterfeit components are common and famous in such inventory. In summary, power electronics circuit designers face many challenges. a) Many applications are mission critical and cannot fail under any circumstances. b) Power electronic circuits must function at both high and low

	temperatures. c) Power electronic circuit components operate at high voltages/currents and the density of the power is high. d) Overall high energy efficiency is desired and critical. e) Higher power converter operating frequencies are necessary to create smaller and lighter modules. f) Fake components are a much bigger issue as compared to low-power components. The examples and solutions for each point a until f must be in essays.
Ways and Means	**Tools** The group must identify the availability/suitable of the selected tools for Power Electronics **Procedure** The group is required to search for the relevant step/information to carry out the study based on the appropriate guideline or manual. **Data Acquisition** All data collected are observed and analysis during the study must be tabulated in the proper format for easy verification and presentation of the technical report.
Results	The group is required to submit a technical report within 3 weeks. In the report. Your task was to provide the examples and solutions to solve each of the challenges above. You should include the introduction, literature review, example and solutions for each point a until f, discussion, conclusion and the references.

NOTE

1) YOUR REPORT (PDF) MUST HAVE COVER PAGE, INTRODUCTION, LITERATURE REVIEW, EXAMPLES AND SOLUTIONS FOR EACH POINT A UNTIL F DISCUSSION, CONCLUSION, REFERENCES.

2) MIN PAGES 15 PAGES. KINDLY COPY THE RUBRICS IN YOUR REPORT.

3) PER GROUP ONLY ONE PERSON UPLOAD TO MOODLE

DUTE DATE: 21 May 2024, 12 PM

RUBRICS

Component Title	Assignment Report				Percentage (%)	100%	
	Score and Descriptors					Weight (%)	Marks
Criteria	Very Good 5	Good 4	Average 3	Poor 2	Very Poor 1		
	CLO1: Demonstrate the understanding in electrical power regulation in various power electronics applications.						
Introduction	Introduction complete and well written; provides all necessary background principles for the experiment	Introduction nearly complete, clear missing some minor points	Introduction is nearly complete, missing some minor points	Some introductory information, but still missing some major points	Very little background information provided or information is incorrect	**10**	
Overall Demonstration and Understanding	Well-planned and well-thought out. Includes title, introduction, statement of main idea, transitions and conclusion**.**	Good overall demostration, includes the main organizational tools.	Average overall demostration, includes the main organizational tools.	There is a sense of demostration, although some of the organizational tools are used weakly or missing	No sense of demonstration and understanding	**20**	
	CLO2: Analyze the models of the power converter control systems.						
Overall Content CLO 2	Exceptionally analyze and argued; ideas are detailed, well-developed, supported with specific evidence & facts, as well as examples and specific details.	Well-analyze, argued; ideas are detailed, developed and supported with evidence mostly specific.	Well-analyze ideas, developed and supported with some evidence	Content is sound and solid; ideas are present & analyze but not particularly developed or supported; some evidence, but usually of a generalized nature.	Content is not good	**15**	
	CLO3: Develop power electronics solutions based on practical problems by applying the basic structures and principles of converters.						
Overall Content CLO 3	Exceptionally wel presented and argued; ideas are detailed, well-developed, supported with specific evidence facts, as well as examples and specific details.	Well-presented, argued; ideas are detailed, developed and supported with evidence mostly specific.	Well-presented ideas, developed and supported with some evidence	Content is sound and solid; ideas are present but not particularly developed or supported; some evidence, but usually of a generalized nature.	Content is not good	**15**	

Conclusions	All important conclusions have been clearly made, student shows good understanding	All important conclusion have been drawn, could be better stated	Some important conclusion have been drawn, could be better stated	Conclusions regarding major points are drawn, but many are miss stated, indicating a lack of understanding	Conclusions missing or missing the important points	**20**	
Appearance and formatting	All sections in order, well-formatted, very readable	All sections in order, formatting generally good but could still be improved	Some sections in order, formatting generally good but could still be improved	Sections in order, contains the minimum allowable amount of handwritten copy, formatting is rough but readable	Sections out of order, too much handwritten copy, sloppy formatting	**20**	
					TOTAL		